AI-Driven Secure Communication Protocols for 6G Networks Using Federated Learning and Lightweight Encryption Techniques

Proposed Workflow

1. Dataset Collection

- 6G mIoT Network Traffic Dataset sourced from Kaggle (colabsss) — specifically designed to represent massive IoT (mIoT) traffic patterns in 6G environments, containing labeled Normal and Attack/Anomaly records representative of 6G-era IoT threat scenarios
- Binary classification target column used: Normal vs. Anomaly/Attack — two-class intrusion detection problem aligned with the dataset's structure
- **https://www.kaggle.com/datasets/colabsss/6g-miot-network-traffic-dataset**

2. Exploratory Data Analysis

- Per-class sample count quantified across both classes — quantifies class imbalance ratio between Normal and Attack records, confirms SMOTE requirement
- Missing value percentage computed per feature — columns exceeding 90% missingness flagged for removal
- Pearson correlation matrix computed across all numerical network traffic features — identifies redundant collinear features before selection
- Statistical distribution analysis performed per feature — informs normalization strategy and identifies high-variance traffic volume features such as packet size, flow duration, and byte counts
- Class imbalance ratio documented — informs SMOTE oversampling configuration per federated client

3. Stratified Dataset Splitting

- Stratified train-validation-test split applied immediately after EDA and before all preprocessing — prevents data leakage by ensuring imputer, scaler, encoder, and feature selector never see validation or test samples during fitting
- Training 70% · Validation 10% · Testing 20% — stratified sampling preserves binary class proportions across all subsets
- All preprocessing fitting operations performed exclusively on training set · transform-only applied to validation and test sets
- Test set held out entirely until final performance evaluation

4. Data Preprocessing (All fitting on training set only · transform applied to validation and test sets)

- High-Missingness Feature Removal: Features exceeding 90% missing rate removed — threshold derived from training statistics only
- Iterative Imputation: Remaining missing values estimated using IterativeImputer with BayesianRidge — each feature modelled as a function of correlated features for accurate reconstruction · fitted on training set only
- Frequency Encoding: Any categorical features such as protocol type or connection state encoded using training-set frequency counts — avoids one-hot dimensionality explosion

- Controlled Outlier Filtering: IQR-based filtering applied only to continuous low-signal features (flow duration, byte ratio features) · Attack-class samples explicitly preserved before and after filtering · thresholds computed from training set only
- Min-Max Normalization: All numerical features scaled to [0,1] — required for stable FT-Transformer feature tokenizer embedding learning · fitted on training set only

5. Feature Selection

- Mutual Information ranking computed between each feature and the binary attack label — scores derived from training set only
- Pearson correlation filtering applied to remove highly collinear features from the ranked list
- Optimal feature subset size determined by evaluating validation binary F1-score across candidate subset sizes {15, 20, 25, 30, 35} — optimal subset selected as the size yielding peak validation F1 · not an arbitrary number
- Selected feature subset applied consistently to training, validation, and test sets

6. Non-IID Federated Client Partitioning

- Pre-processed training data partitioned across five federated clients using Dirichlet distribution with $\alpha = 0.5$ — creates realistic non-IID heterogeneity where each client holds a skewed Normal-to-Attack ratio reflecting its deployment context
- Five clients represent distinct 6G mIoT node types**: IoT sensor cluster · edge compute node · mIoT gateway · smart home hub · 6G base station** each receives a non-uniform distribution of Normal and Attack records matching its realistic traffic profile
- Local data never leaves each client device — privacy enforced by construction across all federated training rounds
- Local validation subset retained per client for early stopping during local training

7. Local SMOTE Class Balancing

- SMOTE applied independently within each client's local training dataset after federated partitioning — synthetic minority Attack samples generated by interpolating between real instances within each client's local feature space
- Applied per client on training partition only — validation and test sets remain untouched across all clients
- Per-client class distribution verified after SMOTE to confirm both Normal and Attack classes are adequately represented for local FT-Transformer training

8. RUN-Based Hyperparameter Optimization

- RUN optimization performed centrally on a representative central validation subset before federated training initialization — ensures one globally optimal hyperparameter configuration applied uniformly across all five clients · avoids per-client inconsistency
- Runge-Kutta Optimizer uses slope functions and Enhancement of Solution Quality mechanism to balance exploration and exploitation · avoids local optima that gradient-free tuning cannot escape
- 30 candidate configurations initialized, each representing a unique FT-Transformer hyperparameter combination

- Search space: learning rate [1e-5–1e-2] · batch size {16, 32, 64, 128} · transformer layers [2–8] · attention heads {2, 4, 8} · embedding dimensions {64, 128, 256} · dropout [0.0–0.5]
- Fitness function: binary F1-score on central validation set — equally penalizes false negatives (missed attacks) and false positives (misclassified normal traffic)
- Best configuration frozen at RUN convergence · federated training initialized with this fixed configuration across all clients

9. Local FT-Transformer Training per Federated Client

- FT-Transformer initialized with RUN-optimal configuration on each of five clients
- Feature Tokenizer converts each selected feature into a learnable dense embedding — CLS token prepended to full feature token sequence as classification anchor · positional encodings omitted since tabular features are unordered
- L stacked Transformer Encoder Blocks each containing multi-head self-attention, feed-forward network, layer normalization, and residual connections — self-attention captures nonlinear interactions between protocol features, byte counts, flag combinations, and timing features for binary attack discrimination
- CLS token final representation passed through a linear projection with Sigmoid activation over 1 output unit — outputs probability of Attack class per traffic record
- Binary cross-entropy loss · Adam optimizer with RUN-selected learning rate · 5 local epochs per federated communication round
- Local early stopping on per-client validation binary F1 — prevents overfitting to skewed local class distribution before aggregation

10. FedProx Global Aggregation

- **FedProx** selected over FedAvg — proximal regularization term $\mu\|w - w\_global\|^2$ added to each client's local objective · penalizes local model from drifting too far from global weights under non-IID Normal/Attack distributions
- Proximal coefficient $\mu = 0.01$ — constrains client drift without suppressing legitimate local adaptation
- Server collects weight updates from all five clients after each round · weighted averaging proportional to local dataset size produces updated global FT-Transformer
- Global model redistributed to all clients for next local training round — no raw traffic data ever transmitted to server
- 100 communication rounds executed · convergence monitored by global validation binary F1 and global training loss · early stopping if improvement falls below threshold for 10 consecutive rounds
- Client drift reduction reported per round — quantifies FedProx advantage over FedAvg under same non-IID conditions · global loss convergence curve reported across all 100 rounds

11. Binary Attack Detection and Traffic Routing

- Trained global FT-Transformer deployed at 6G network inference layer — incoming traffic preprocessed using frozen pipeline before inference
- Single forward pass-through feature tokenizer · encoder blocks · CLS Sigmoid head produces binary Attack probability per traffic record

- Per-class confidence threshold calibrated from validation precision-recall curve by maximizing binary F1 — avoids fixed 0.5 threshold which underperforms under imbalanced Attack/Normal priors
- Traffic routed based on predicted class:

Predicted Class	***Action***
Normal	Forwarded to Ascon-128 encryption pipeline for secure transmission
Attack / Anomaly	Blocked · source features · timestamp · confidence score logged · alert triggered

12. Ascon-128 Lightweight Encryption

- Ascon-128 AEAD applied to Normal-classified traffic — NIST-standardized lightweight cryptographic algorithm for resource-constrained IoT and 6G mIoT environments · selected over AES-GCM due to lower computation, memory, and power requirements
- Provides confidentiality, integrity, and authenticity in a single cryptographic operation — no separate MAC computation needed
- Encryption inputs: plaintext payload · 128-bit secret key · 96-bit unique nonce (never reused) · associated data (packet headers authenticated but not encrypted)
- 320-bit internal state processed through initialization · associated data absorption · plaintext encryption — output is ciphertext of identical length plus 128-bit authentication tag

13. Secure Transmission and Ascon Decryption

- Encrypted packets routed through 6G network — opaque to any intermediate relay lacking the session key
- Receiving node verifies 128-bit authentication tag before any plaintext release — tampered or replayed packets discarded with integrity violation alert · verify-before-decrypt prevents malicious payload processing
- On successful tag verification, ciphertext decrypted using shared key and transmitted nonce — same 320-bit permutation as encryption
- Nonce uniqueness enforced using monotonically incrementing counter per device per session

14. Performance Evaluation

Binary Classification Metrics: Accuracy · Precision · Recall · F1-Score · ROC-AUC · Matthews Correlation Coefficient (MCC) · Confusion Matrix (2×2) · False Positive Rate · False Negative Rate

Federated Learning Metrics:

- Communication cost (MB/round) · Convergence rate (rounds to target binary F1) · Training time per round · Client drift reduction (FedProx vs FedAvg) · Global loss convergence curve across 100 rounds

Encryption Metrics (Secure Communication) :(graphs)

- Encryption throughput (MB/s) · Decryption throughput (MB/s) · Per-packet encryption latency (μs) · Per-packet decryption latency (μs) · Memory footprint (KB) · Energy consumption per encryption operation (mJ) · Energy consumption per decryption operation (mJ) · Authentication tag verification time (μs) · Nonce generation overhead (μs) · Comparison against AES-128-GCM across all above metrics

Baselines:

- **Centralized FT-Transformer · FedAvg + FT-Transformer · FedProx + LSTM · FedProx + MLP · Proposed RUN-FTTransformer-FedProx-Ascon framework**

Ablation Study: Baseline FT-Transformer → add SMOTE → add feature selection → add RUN optimization → add non-IID FL with FedProx → add Ascon-128 encryption → full framework

Additional Plots: RUN convergence curve · Global binary F1 vs communication rounds · Global loss convergence curve · Precision-recall curve (Attack class) · ROC curve · Optimal hyperparameter table · SMOTE per-client class distribution · 2×2 confusion matrix heatmap · Feature subset size vs validation binary F1 (justifies optimal feature count selection) · Ascon-128 vs AES-128-GCM throughput and latency bar charts · Energy per operation comparison chart